MATH 110 — Week 7 — Solutions

Notes Chapters 12–13; Homework 7

HW7 P1.

Idea. Compute the first polynomial relation, then check whether its factors split and are distinct.

1. $A^2 = -I$, and A is not scalar over either field. Thus $m_A = x^2 + 1$, irreducible over $\mathbb{R}$, but $(x - i)(x + i)$ over $\mathbb{C}$. Hence A is diagonalizable only over $\mathbb{C}$.
2. $(B - I)^2 = 0$ but $B - I \neq 0$. Thus $m_B = (x - 1)^2$, so B is not diagonalizable over $\mathbb{R}$.

HW7 P2.

Idea. The iterates of x^3 detect every possible lower-degree relation.

1. $D^4 = 0$, while $x^3, Dx^3, D^2x^3, D^3x^3 = x^3, 3x^2, 6x, 6$ are independent. Thus $m_D = x^4$.
2. Its repeated root rules out diagonalizability by Proposition 68.

HW7 P3.

Idea. A nonzero constant term allows the operator relation to be rearranged into an inverse.

1. If $a_0 = 0$, write $m_T = xq$; invertibility and $Tq(T) = 0$ imply $q(T) = 0$, contradicting minimality.
2. Rearranging gives $T^{-1} = -a_0^{-1}(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I)$. For $d = 0$ on the zero space, $T^{-1} = I = 0$.
3. In (c), $m_T = (x-1)^2(x^2+3x+4)$. Thus $T^{-1} = (-T^3 - T^2 + T + 5I)/4$. Its eigenvalues are $1, (-3+i\sqrt{7})/2, (-3-i\sqrt{7})/2$, by Proposition 66 and the quadratic formula.
4. The repeated root 1 in the minimal polynomial rules out diagonalizability, by Proposition 68.

HW7 P4 (optional).

Idea. Affine substitution preserves annihilation and degree; reverse it to exclude smaller relations.

1. Put $S = T - \lambda I$ and $q(x) = m_T(x + \lambda)$. Then q is monic of the same degree as m_T and $q(S) = m_T(T) = 0$.
2. If a nonzero polynomial h of smaller degree annihilated S, then $h(x - \lambda)$ would annihilate T with smaller degree, impossible. Hence $m_S = q$.
3. For $R = cT$, $q(x) = c^d m_T(x/c)$ is monic of degree d and $q(R) = 0$. A smaller annihilator h of R would give the smaller annihilator $h(cx)$ of T. Thus $m_R = q$.

HW7 P5.

Idea. The root set and the degree bound leave just two monic possibilities.

1. Complex factorization and Proposition 66 imply $m_A = (x-\lambda)^d$. Since the space is nonzero, $d \geq 1$, and Proposition 67 gives $d \leq 2$.
2. The relation $m_A = x - \lambda$ is equivalent to $A - \lambda I = 0$. Proposition 68 gives diagonalizability exactly in this first case.

HW7 P6.

Idea. The repeated diagonal value has only one eigenvector direction.

1. Triangular form gives eigenvalues 2 and 3. Solving $(C-2I)v=0$ gives $E(2,C)=Fe_1$; solving $(C-3I)v=0$ gives $E(3,C)=Fe_3$ (over $\mathbb{R}$ or $\mathbb{C}$).
2. $(C-2I)(C-3I)=\left(\begin{smallmatrix}0&-1&0\\0&0&0\\0&0&0\end{smallmatrix}\right)\neq 0$, whereas $(C-2I)^2(C-3I)=0$. Thus $m_C=(x-2)^2(x-3)$.
3. The repeated root rules out diagonalizability. Equivalently the two eigenspace dimensions sum to 2, less than 3.

HW7 P7 (optional).

Idea. Interpolate the desired scalar action on each eigenspace.

1. Set $p_j(x)=\prod_{k\neq j}(x-\lambda_k)/(\lambda_j-\lambda_k)$. All denominators are nonzero, and substitution gives the required values.
2. Proposition 68 gives the direct decomposition $v=\sum_k v_k$ with $v_k\in E(\lambda_k,T)$. Week 6 evaluation on eigenvectors gives $p_j(T)v_k=p_j(\lambda_k)v_k$.
3. Therefore $p_j(T)v=v_j$, which is precisely the stated projection. It is identity on its range, zero on the other summands, and satisfies $p_j(T)^2=p_j(T)$.

Exam practice, Problem 6.

Idea. Use the same first-dependence and remainder arguments on one vector.

1. If $v=0$, take $p_v=1$; it is the unique monic degree-0 polynomial and divides every polynomial. Otherwise the finite-dimensional dependence of $v,Tv,\dots$ gives a nonzero annihilating polynomial for v. Choose minimal degree and normalize it to be monic.
2. Two distinct monic choices of the same degree would have nonzero lower-degree difference annihilating v, a contradiction; hence p_v is unique.
3. Divide $m_T=p_vh+r$ with $\deg r<\deg p_v$ or $r=0$. Evaluation on v gives $r(T)v=0$ because $p_v(T)$ commutes with $h(T)$. Minimality forces $r=0$, so $p_v\mid m_T$.